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Novel Stereo-Induction Pattern in Pudovik Addition/*Phospha*-Brook Rearrangement Towards Chiral Trisubstituted Allenes

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Dedicated to Professor Xiaoming Feng and Professor Yong Tang on the occasion of their respective 60th birthdays.

Abstract: Despite the significance of chiral allene skeletons in catalysis, organic synthesis and medicinal chemistry et al., there is a scarcity of reports on axially chiral allenyl phosphorus compounds. Here, we disclosed an efficient and straightforward cascade reaction between ethynyl ketones and phosphine oxides, resulting in a broad array of trisubstituted allenes incorporating a phosphorus moiety in high yields with excellent stereoselectivities facilitated by peptide-mimic phosphonium salt (PPS) catalysis, Additionally, comprehensive series of mechanistic experiments have been conducted to elucidate that this cascade reaction proceeds via an asymmetric Pudovik addition reaction followed by a subsequent *phospha*-Brook rearrangement that occurs concomitantly with kinetic resolution, representing a stereospecific rearrangement and protonation process facilitating central-to-axial chirality transfer in a cascade manner. We anticipate that our research will pave the way for a promising exploration of novel stereo-induction pattern in the Pudovik addition/*phospha*-Brook rearrangement cascade reaction.

Introduction

The pursuit of asymmetric catalysis for generating diverse and practical enantioenriched materials with axial chirality has been a continuous endeavor in modern organic chemistry, resulting in numerous significant advancements. Concurrently, chiral allenes, characterized by two cumulated carbon-carbon double bonds, constitute one of the most significant categories. They serve as crucial building blocks in various bioactive molecules, pharmaceutical compounds, functional materials, and organic synthesis.[1] Consequently, general, concise, and straightforward catalytic asymmetric methodologies for synthesizing these allenylic molecules with axial chirality have been extensively developed in recent decades. Many of these methodologies rely on the isomerization of propargyl derivatives.[2] Recently, remarkable advancements have been achieved in catalyst-controlled asymmetric synthesis of chiral allenes through both transition-metal (TM) catalysis and organocatalysis (Figure 1A). On the former, the representative contributions mainly come from the research groups of Ma,[3] Zhang,[4] Liu,[5] Guo[6] and Wang[7] et al. particularly by utilizing propargyl derivatives to synthesize optically pure allenylic molecules under metal-catalytic conditions. In terms of organocatalysis, a less explored area, some electron-deficient systems, such as enynes,[8] alkynoates,[9] imines,[10] and other reactants[11] formed in situ from functionalized propargyl alcohols under acidic conditions have been utilized as acceptors in enantioselective addition reactions to furnish axially chiral allenes. Despite these efforts, most attempts to produce allene molecules have encountered challenges such as limited reaction partners with broad substituents, elusive mechanisms, and unclear stereoinduction patterns. Therefore, there remains an urgent need to develop novel methodologies for constructing chiral allenes from simple and readily available reactants under transition-metal-free conditions.

Conjugated ynones, also known as α, β-ynones characterized by a carbonyl group adjacent to an alkynyl group, have drawn significant attention from chemists due to their attractive chemoreactivity. They participate not only in cycloaddition reactions but also in addition reactions with numerous nucleophiles (Nu) owing to the electrophilicity and polarization of the triple bond (Figure 1B).[12] It is noteworthy that the reactivity of simple ynones bearing two electron-neutral groups (ENG) is primarily focused on the latter transformations involving attack at the β-carbon (i.e.

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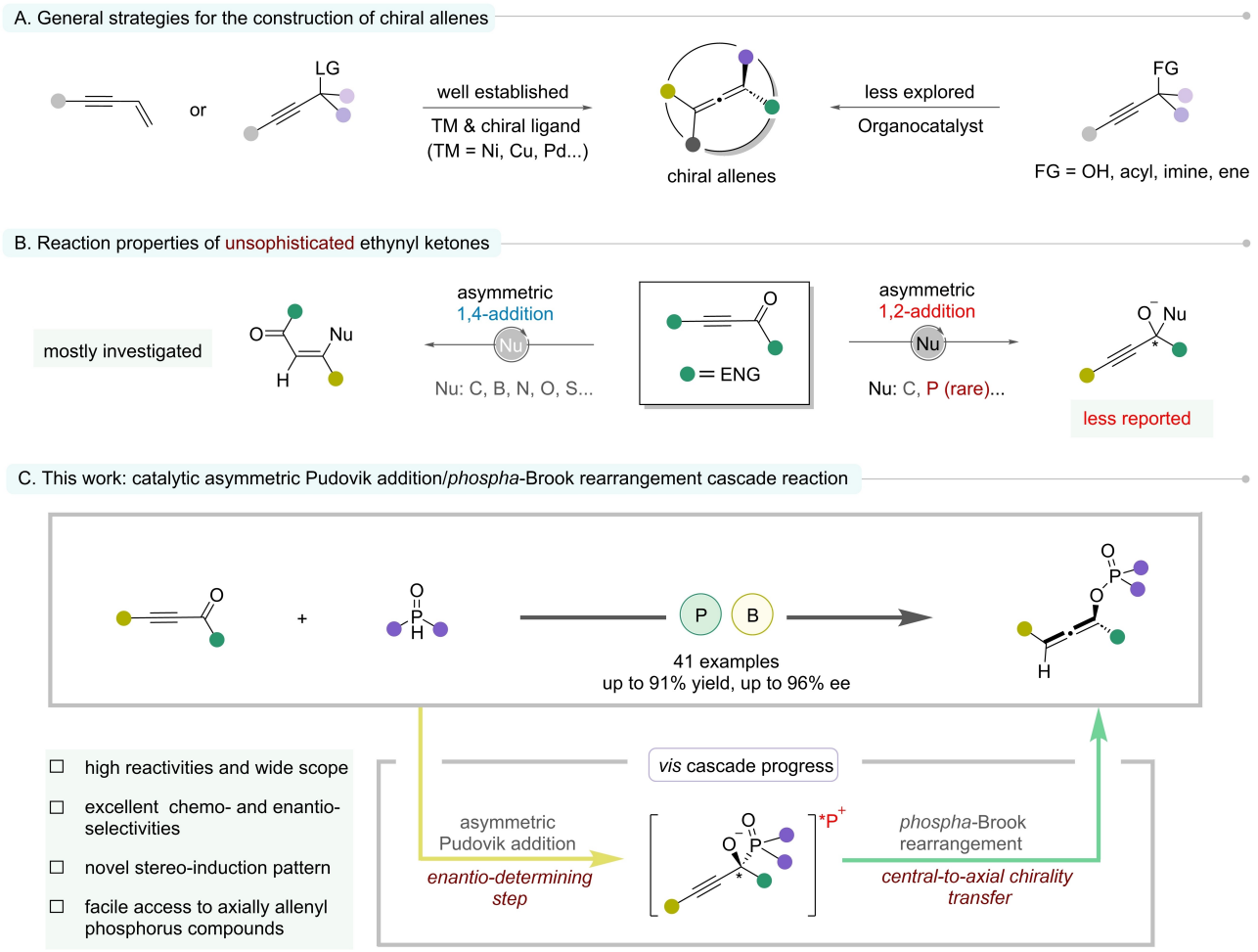


Figure 1. Catalytic asymmetric cascade reaction of ynones towards chiral allenes.

1,4-addition), along with more straightforward reactions, although partly initiated by addition of the carbonyl group (i.e. 1,2-addition), and generally differing in the nature of nucleophiles. In well-established examples, soft nucleophiles,[13] including boron,[14] carbon,[15] nitrogen,[16] chalcogen[17] and radical,[18] are inclined to directly attack the unactivated α, β-ynones at the β-position in a Michael-like manner, resulting in the formation of unsaturated alkenones as major products; conversely, 1,2-addition to the carbonyl group occurs when hard nucleophiles are employed, typically restricted to reactants, such as organometallic reagents,[19] TMSCN[20] and $TMSCF_3$,[21] which display significant nucleophilic character; moreover, Ooi et al. achieved enantioselective hydrophosphonylation of ynones through 1,2-addition, facilitated by the pronounced nucleophilicity of phosphite species via chiral *tetra*-aminophosphonium catalysis.[22] Of note, compared to structurally diverse chiral compounds, such as chiral quaternary carbon centers, chiral spirocyclic molecules,[23] chiral axially chiral compounds[17e] and others, accessed through asymmetric 1,4-addition or cycloaddition reactions involving ynones, chiral alcohols represent the predominant product obtained through catalytic enantioselective 1,2-addition reactions of ynones.[19–22]

Thus, there is a compelling drive and appeal to delve deeper into novel asymmetric catalytic methodologies, employing the 1,2-addition of unactivated ynones for the synthesis of chiral molecules.

However, despite its appeal as mentioned above, some intrinsic challenges associated with this hypothesis can be anticipated. First, the absence of affiliated activation groups renders ynone carbonyls insufficiently electrophilic, which hinders their reactivity with hetero-nucleophiles. Moreover, to our knowledge, in the realm of enantioselective 1,2-addition reactions to unsophisticated ynones, the majority of reports have concentrated on synthesizing chiral alcohols but neglecting other chiral products. Quite recently, Feng[24] reported an efficient catalytic system involving an *N,N'*-dioxide/Sc(III) complex for the 1,2-addition of α-alkynylketoamides with diarylphosphine oxides, followed by a *phospha*-Brook rearrangement,[25] and their mechanistic results revealed that the amide group played a crucial role not only in activating the carbonyl but also in facilitating subsequent key step of enantioselective protonation via enolization. Therefore, there is still promise in exploring the reactivity of simple acetylenic ketones and gaining a comprehensive understanding of the mechanism. Drawing

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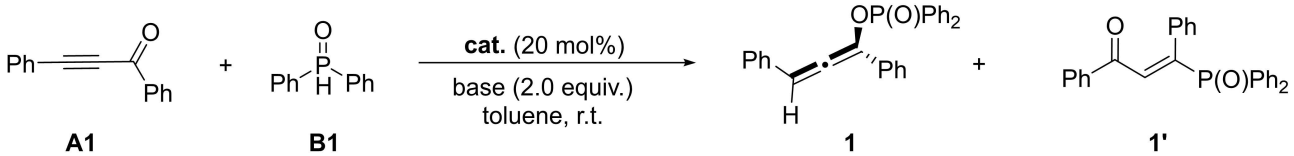
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from our extensive investigations on peptide-mimic phosphonium salt (PPS) catalysis,[26] we recognize that these highly tunable multifunctional phosphonium halide catalysts, particularly those that possess semi-enclosed cavities and also have fascinating electrostatic potentials (ESPs) with an electropositive region, hold promise for addressing reactivity and selectivity issues in 1,2-addition-triggering cascade reactions of simple ynones with hetero-nucleophiles (Figure 1C). Here, we present an efficient cascade reaction involving Pudovik addition/*phospha*-Brook rearrangement of simple ynones with phosphorus species, resulting in high yields and excellent enantioselectivities of axially chiral trisubstituted allenes. Mechanistic findings demonstrate that while the initial 1,2-addition adeptly controls stereoselectivity, the subsequent rearrangement, which includes a central-to-axial chirality conversion, also increases the demands on the chiral catalysts.

Results and Discussion

To validate our hypothesis, we synthesized the starting materials 1,3-diphenylprop-2-yn-1-one **A1** and diphenylphosphine oxide **B1**. The model reaction was then conducted using **A1** and **B1** as substrates in toluene with Cs_2CO_3 at room temperature employing PPSs as catalysts (Table 1). While the desired allenes were smoothly obtained with PPS catalyst, the by-product, (*E*)-3-(diphenylphosphoryl)-1,3-diphenylprop-2-en-1-one **1′**, resulting from a 1,4-addition process, was isolated in 76% yield. In contrast, under PPS catalytic conditions, the yield of the by-product dramatically decreased to less than 5%, with the expected allene becoming the main product, highlighting the crucial role of PPS in controlling the chemoselectivity of this cascade reaction (entries 1 and 2). Next, a series of PPS catalysts were employed in the model reaction. To our delight, all of them promoted the cascade reaction effectively. When *L*–Val-derived phosphonium salts **P1**-**P4** were used, the desired products were obtained in high yields with unsatisfactory ee values (entries 2–5). While the R^1 of PPS was replaced by *tert*-butyl group with great steric hindrance (**P5**–**P6**), the corresponding product was formed in higher isolated yields with increased ee values under such catalyst systems (entries 6 and 7). Building upon these pivotal findings, additional phosphonium bromide salts derived from *L*-*tert*-leucine or *L*-threonine and featuring diverse silicon-containing groups were utilized (entries 8–14). It emerged that catalyst **P11** proved to be the most efficacious, yielding the expected allenylic products with 88% yield and 71% ee (entry 12). Employing the optimal catalyst **P11**, the effect of other reaction conditions such as solvent, base, reaction temperature and chiral catalyst loading was evaluated (entries 15–21). These investigations revealed that optimal results (87% yield, 93% ee) were attained by

Table 1: Reaction optimization.[a]



A1 **B1** **1** **1'**

entry	cat.	base	time (h)	yield of **1** (%)	yield of **1'** (%)	ee (%)
1	none	Cs_2CO_3	3	18	76	–
2	**P1**	Cs_2CO_3	3	78	<5	22
3	**P2**	Cs_2CO_3	3	79	<5	28
4	**P3**	Cs_2CO_3	3	82	<5	40
5	**P4**	Cs_2CO_3	3	87	<5	43
6	**P5**	Cs_2CO_3	3	87	<5	50
7	**P6**	Cs_2CO_3	3	88	<5	57
8	**P7**	Cs_2CO_3	3	87	<5	56
9	**P8**	Cs_2CO_3	3	87	<5	65
10	**P9**	Cs_2CO_3	3	89	<5	59
11	**P10**	Cs_2CO_3	3	88	<5	67
12	**P11**	Cs_2CO_3	3	88	<5	71
13	**P12**	Cs_2CO_3	3	87	<5	68
14	**P13**	Cs_2CO_3	3	85	<5	45
15	**P11**	K_3PO_4	3	85	<5	59
16	**P11**	LiOH	3	88	<5	73
17	**P11**	KOH	2	90	<5	61
18	**P11**	$K_3PO_4•3H_2O$	3	86	<5	75
19[b]	**P11**	$K_3PO_4•3H_2O$	3	89	<5	80
20[b,c]	**P11**	$K_3PO_4•3H_2O$	36	87	<5	93
21[b,c,d]	**P11**	$K_3PO_4•3H_2O$	48	87	<5	93

P1, **P2**: PPh_2MeI^-, NH, NHBoc

P3: R^1 = ^{i}Pr, R^2 = ^{i}Pr
P4: R^1 = ^{i}Pr, R^2 = ^{t}Bu
P5: R^1 = ^{t}Bu, R^2 = ^{i}Pr
P6: R^1 = ^{t}Bu, R^2 = ^{t}Bu

P13

R^2 = ^{t}Bu, R^3 = H:
R^1 = TBDPS (**P7**)
R^1 = TIPS (**P8**), R^1 = TBS (**P9**)
R^1 = TIPS, R^2 = ^{t}Bu:
R^3 = Me (**P10**), R^3 = CF_3 (**P11**)
P12: R^1 = TIPS, R^2 = Ph, R^3 = CF_3

[a] Conditions: reactants **A1** (0.20 mmol) and **B1** (0.30 mmol), base (0.4 mmol) and chiral catalyst (20 mol%) in toluene (5.0 mL) at room temperature. All ee values were determined by chiral HPLC. Isolated yields. [b] With $K_3PO_4·3H_2O$ (8.0 equiv.). [c] At −40 °C for 36 h. [d] With **P11** (10 mol%). TBDPS = *tert*-butyldiphenylsilyl, TIPS = triisopropylsilyl, TBS = *tert*-butyldimethylsilyl.

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executing the model reaction at −40 °C, employing 10 mol % **P11**, and using $K_3PO_4 \cdot 3H_2O$ (8.0 equiv.) in toluene for 48 hours (see Tables S1–S4 in Supporting Information for more details).

With optimized reaction conditions established, our attention turned to exploring the scope of this PPS-catalyzed asymmetric cascade reaction involving Pudovik addition/*phospha*-Brook rearrangement. Initially, we evaluated the range of ethynyl ketones using diphenylphosphine oxide **B1** as the co-reactant (Figure 2). Remarkably, when the phenyl rings directly attached to the ethynyl group of the ethynyl ketones carried electron-donating, neutral, or electron-withdrawing substituents, these substrates reacted smoothly, yielding axially chiral allenyl phosphine oxides (**1**–**8**) in high yields (85–91 %) and with excellent enantioselectivities (90–93 % ee). Despite significant steric hindrance, the cascade reaction proceeded smoothly, yielding products (**9**–**16**) with high reactivities and excellent ee values (up to 96 % ee) using phosphonium bromide salt **P12**. Moreover, a variety of aromatic rings attached directly to the carbonyl group of ethynyl ketones, featuring electron-donating, neutral, or electron-withdrawing substituents, were also compatible with the standard reaction conditions, providing the desired products (**17**–**27**) in excellent yields (83–91 %) and enantioselectivities (up to 96 % ee).

Taking into account the impact of substituents on phosphine oxides, subsequently, we directed our focus towards exploring the scope of phosphine oxides and carried out reactions with **A12** under the optimized condition. Remarkably, a wide variety of phosphine oxides bearing electron-donating, neutral, or electron-withdrawing groups in either the *meta*- or *para*-position of the phenyl ring reacted effectively, yielding the desired products (**28**–**35**) in satisfactory yields with excellent enantiomeric excess values (88–94 %). Moreover, phosphine oxide derivatives were also proved as suitable substrates for this protocol, providing the desired products (**36** and **37**) with excellent enantioselectivities. Additionally, when fused-ring substituted phosphine oxides (**B38** and **B39**) were employed, axially chiral allenylic products were obtained successfully in good yields and enantioselectivities. To expand the range of allenylic products, we further explored phosphine oxide substituted with heteroaryl group and obtained compound **40** in 83 % yield with 88 % ee. Of note, allenylphosphonate **41** were also successfully formed in good yield with moderate enantioselectivity (72 % ee). The absolute configuration of product **14** was determined by X-ray crystallographic analysis (CCDC 2252444); and other allenylic products were assigned by analogy (see Table S5 in Supporting Information for details).[27]

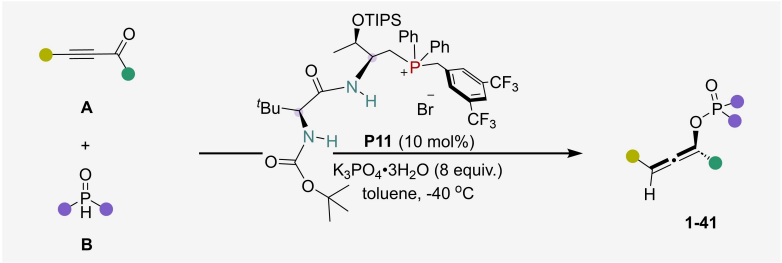
After developing such efficient and practical sequential strategy for synthesizing chiral phosphorus-containing allenes, we aimed to validate its scalability (Figure 3). Scaled-up preparation of compounds **1** and **12** were carried out under standard conditions with KOH (0.5 equiv.) instead of $K_3PO_4 \cdot 3H_2O$ (8.0 equiv.), yielding the desired products with maintained isolated yields and similar enantioselectivities in a significantly shorter time. To further expand the applicability of this protocol, several bioactive molecule scaffolds were incorporated into compound **A14**, thus yielding compounds **A42**–**A44**. All of them performed very well under the standard condition b, providing the corresponding chiral allenes in good yields with satisfying stereoselectivities (up to 96 % ee). Additionally, upon treatment with *tetra-n*-butylammonium fluoride (TBAF) at −78 °C for 10 minutes, the trimethylsilyl (TMS) group of compound **16** was readily removed, yielding compound **45** in moderate yield. The coupling reaction between the alkyne group of compound **45** and iodobenzene successfully occurred and yielded allene **46**.

To elucidate the mechanism of this cascade system, we performed a series of control experiments (Figure 4). Initially, the reaction pathway was monitored, and it was evident that neither intermediates nor allene products were obtained within the first 5 hours. The intermediate gradually accumulated thereafter and subsequently converted into the allenylic product **41** (Figure 4a and entry 1 in Figure 4c). Meanwhile, as the starting materials were consumed, the enantiomeric excess value of **41** remained at 72 %. To understand the key step of asymmetric induction in this cascade process, we performed additional control experiments (Figure 4b and 4c). While racemic intermediate (*rac*-**Int**) was performed under standard condition c, the enantiomeric excess (ee) of the remaining intermediate (*S*-**Int**) increased gradually, suggesting that the rearrangement of the intermediate likely involves a kinetic resolution process. Next, optically pure intermediates were also tested under standard condition c to confirm the predominant conformation of **Int** in the subsequent step. When (*R*)-**Int** was investigated (Figure 4b and entry 3 in Figure 4c), the transformation was completed in about 18 hours, yielding (*S*)-**41** with almost equal ee value, indicating that (*R*)-**Int** was the favorable and matched species in the second kinetic resolution process. Conversely, the reaction of (*S*)-**Int**, whose absolute configuration was determined by electronic circular dichroism (ECD) spectra calculated via the time-dependent-density functional theory (TD-DFT) method (see Figure S6 in Supporting Information for details), took much longer time to yield (*R*)-**41** in reduced yield, while the ee value of (*R*)-**41** started at 53 % and gradually increased to 69 % during the reaction (Figure 4b and entry 4 in Figure 4c). When we introduced optically pure (*R*)-**Int** and (*S*)-**Int** into the reaction catalyzed by (*rac*)-**P13**, the corresponding products via rearrangement and protonation were observed, resulting in a reduction in the ee value of each, indicating that the entire progress of this cascade reaction involved enantioselective 1,2-addition followed by stereoselective rearrangement and protonation (entries 5 and 6 in Figure 4c). Of note, almost no products were obtained without **P13** (entries 7 and 8 in Figure 4c). Besides, the relationship between the ee value of catalyst **P12** and product **12** exhibited a significantly non-linear effect (due to additive Horeau amplification), further confirming the involvement of a kinetic resolution process controlled by catalyst (see Figure S15 in Supporting Information for details).

Based on our prior investigations into peptide-mimic phosphonium salt-catalyzed transformations,[26f] it is evident

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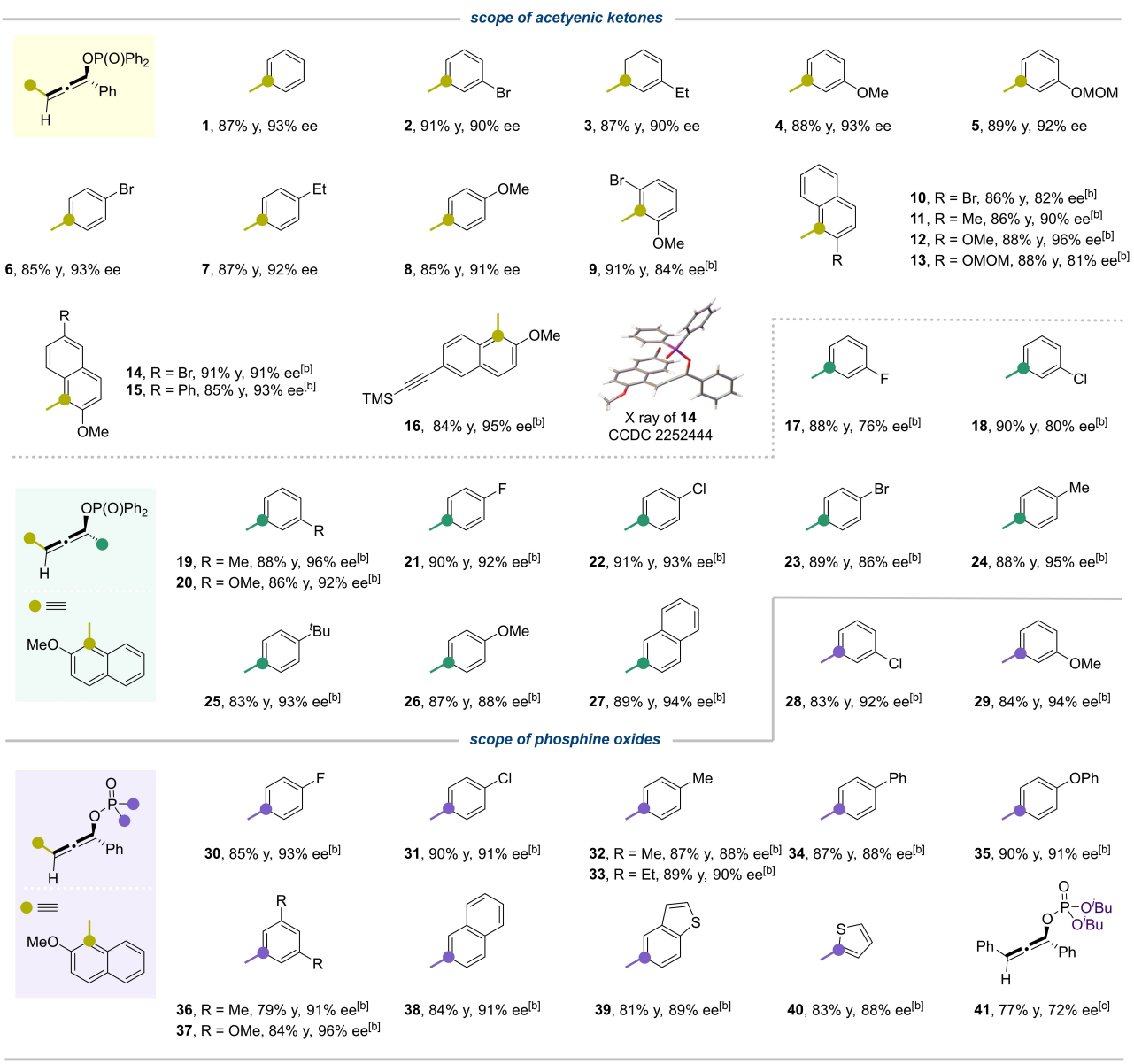


Figure 2. Substrate scope.[a] [a] Standard condition a: Unless other noticed, reactions were carried out with **A** (0.20 mmol), **B** (0.3 mmol), $K_3PO_4 \cdot 3H_2O$ (1.60 mmol) and **P11** (10 mol%) in toluene (5.0 mL) at −40 °C for 2–3 days. Isolated yields based on **A** were given. All ee values were determined by chiral HPLC analysis. [b] Standard condition b: Carried out with **P12** (10 mol%) and Cs_2CO_3 (1.60 mmol) at −30 °C for 2–3 days. [c] Standard condition c: Carried out with **P13** (10 mol%) and $K_3PO_4 \cdot 3H_2O$ (1.60 mmol) at −50 °C for 4 days.

that hydrogen-bonding interactions, coupled with ion-pair activation, significantly contribute to asymmetric induction. Consequently, ^{1}H NMR titration experiments were conducted between **P11** and substrates to elucidate the non-covalent interaction pattern (Figure 5a). The chemical shifts of the amide-NH and benzyl C–H signals of **P11** are readily discernible. Additionally, a Job-plot curve was constructed (Figure 5b), indicating a 1:1 bonding pattern between **P11**

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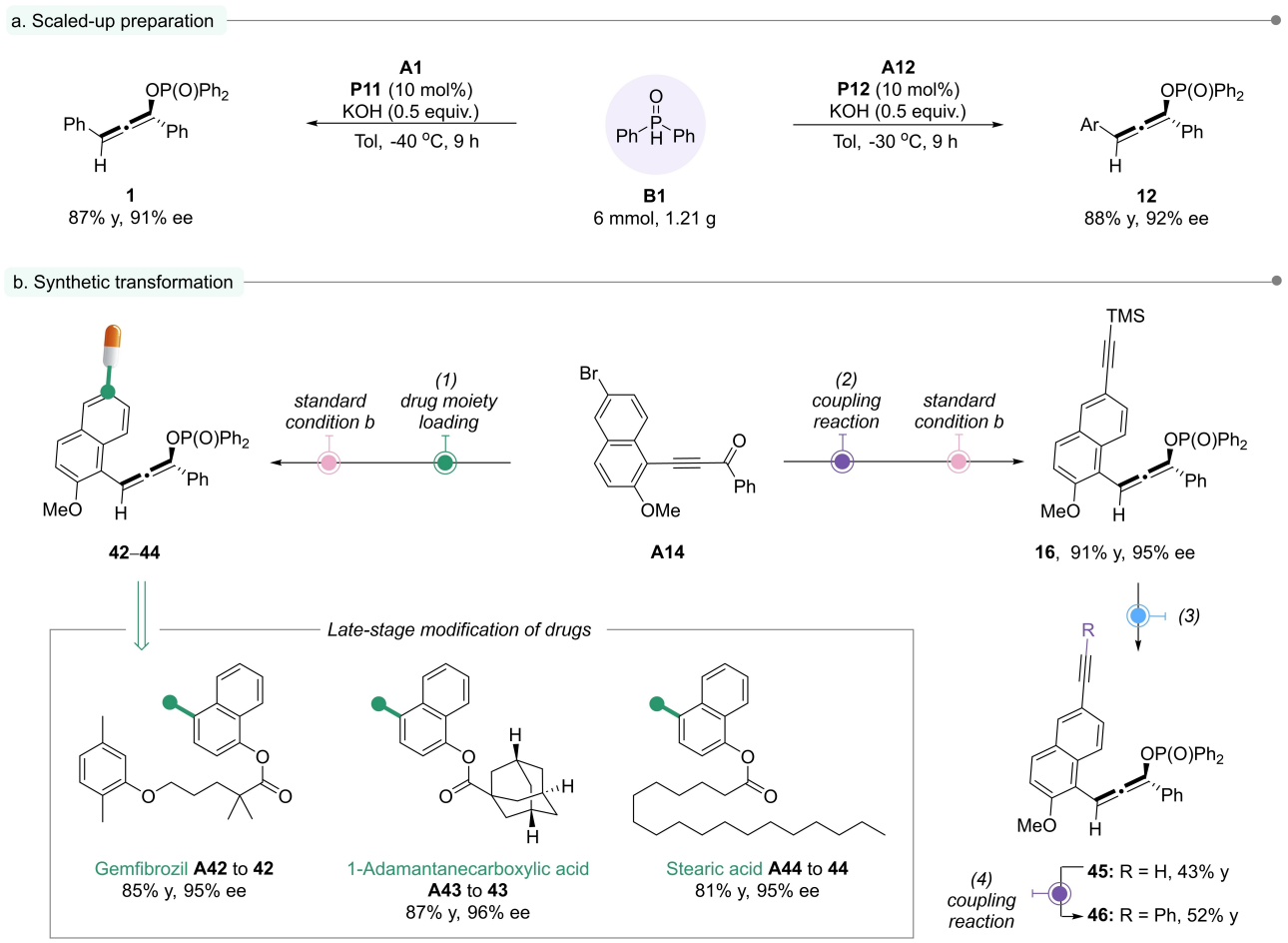


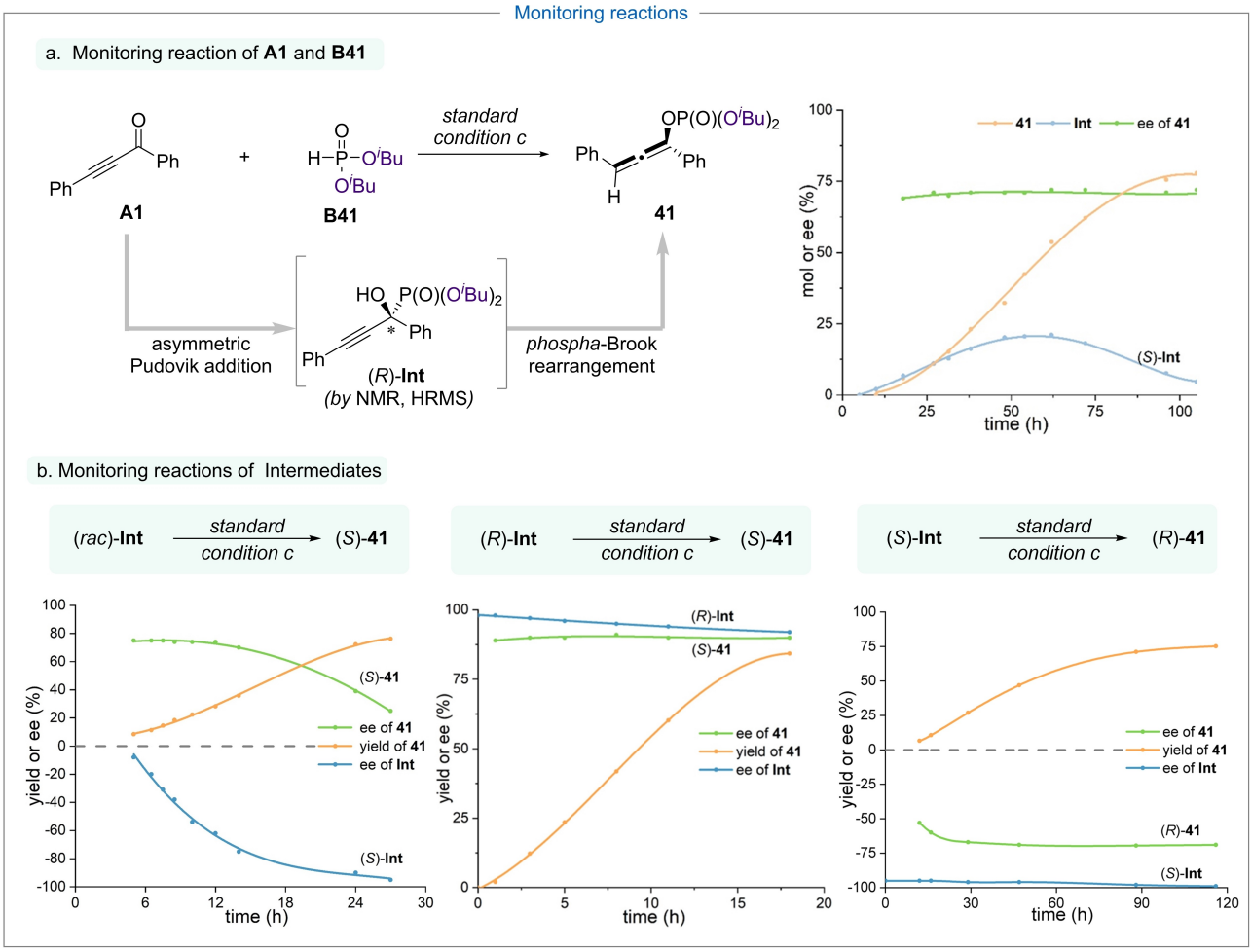
Figure 3. Scaled-up preparation and synthetic transformation. Conditions: (1) Naphthylboronic acid containing drug moiety (1.2 equiv.), $Pd(PPh_3)_4$ (5 mol%), K_2CO_3 (2.0 equiv.), THF:H_2O = 1:1 (v/v), 80 °C, under N_2. (2) Ethynyltrimethylsilane (1.2 equiv.), $Pd(PPh_3)_2Cl_2$ (2 mol%), CuI (5 mol%), Et_3N (1.2 equiv.), 70 °C, under N_2. (3) TBAF (40 mol%), THF, −78 °C, under N_2, 10 min. (4) PhI (1.2 equiv.), $Pd(PPh_3)_2Cl_2$ (2 mol%), CuI (5 mol%), Et_3N (1.2 equiv.), 45 °C, under N_2.

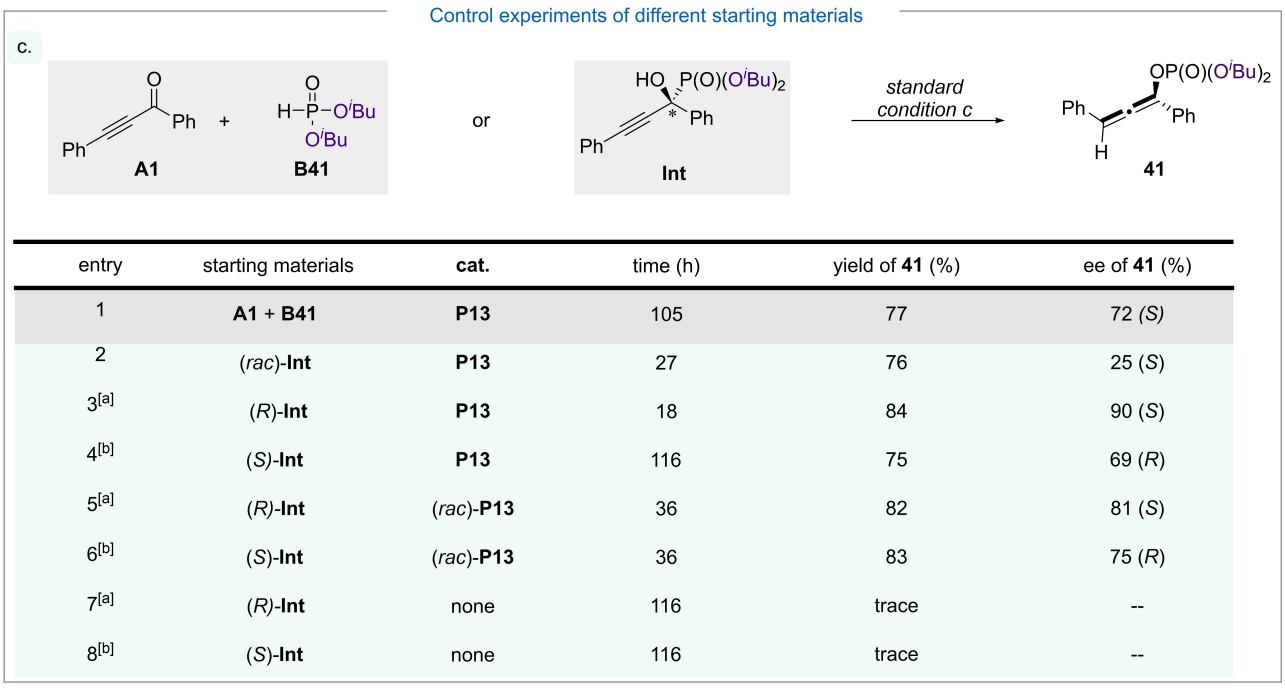
and each substrate. Moreover, catalysts **P12-1** and **P12-2**, with each amide-NH protected by a methyl group, were synthesized and utilized in the catalytic reaction under standard conditions, respectively replacing **P12**. Unsurprisingly, both the yield and ee value dropped rapidly with **P12-1** and **P12-2**. When polar methanol replaced toluene as the solvent, only a small amount of racemate was formed in the system. Finally, two deuterium-labeling experiments were conducted to investigate the source of hydrogen in the chiral allene products. The model reaction was conducted with D_2O (10.0 equiv.) or $DP(O)Ph_2$. A 66% deuterium ratio was obtained from the system containing D_2O, while a 0% deuterium ratio was observed in the $DP(O)Ph_2$ system, suggesting that water is the source of the proton in the system. These mechanistic studies elucidated a clear and elaborate transition state model as **TS-I**-*R*.

Ultimately, a plausible mechanistic cycle delineating a novel stereoinduction pattern in the Pudovik addition/*phospha*-Brook rearrangement cascade toward chiral trisubstituted allenes catalyzed by PPS was outlined (Figure 6a). Generally, base-triggered deprotonation of **B1** resulted in the formation of the enantioenriched alkoxide (*R*)-**Int** through nucleophilic attack on **A1** via **Ts-I**-*R*, which was identified as the enantiodetermining step. Subsequently, a stereospecific *phospha*-Brook rearrangement occurred via **TS-II**. Importantly, a kinetic resolution process was proposed to occur when the matched **TS-II**-*R* underwent successful protonation, yielding product **1**, whereas the protonation of the mismatched one proceeded slowly. This was confirmed through a series of mechanism control experiments. Furthermore, density functional theory (DFT) calculations using the Gaussian 09 program[28] (see Supporting Information for more details) were conducted to gain deeper insight into the enantio-determining step (Figure 6b). The positively charged catalyst **P11** activated two substrates (**A1** and **B1**) through ion-pair interaction as well as multiple hydrogen-bondings. Furthermore, non-covalent interactions revealed the presence of more stabilizing hydrogen-bonding interaction between the CF_3 group of **P11** and the hydrogen atoms in the phenyl group of **A1**, as well as CH···π interactions. Consequently, the △G of **TS-I**-*R* was 2.9 kcal mol^{-1} lower than that of **TS-I**-*S*, thus providing (*R*)-**Int** as key intermediate for the following rearrangement.

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entry	starting materials	cat.	time (h)	yield of **41** (%)	ee of **41** (%)
1	**A1** + **B41**	**P13**	105	77	72 (*S*)
2	(*rac*)-**Int**	**P13**	27	76	25 (*S*)
3[a]	(*R*)-**Int**	**P13**	18	84	90 (*S*)
4[b]	(*S*)-**Int**	**P13**	116	75	69 (*R*)
5[a]	(*R*)-**Int**	(*rac*)-**P13**	36	82	81 (*S*)
6[b]	(*S*)-**Int**	(*rac*)-**P13**	36	83	75 (*R*)
7[a]	(*R*)-**Int**	none	116	trace	--
8[b]	(*S*)-**Int**	none	116	trace	--

Figure 4. Control experiments. [a] With 98% ee of (*R*)-**Int**; [b] With 95% ee of (*S*)-**Int**.

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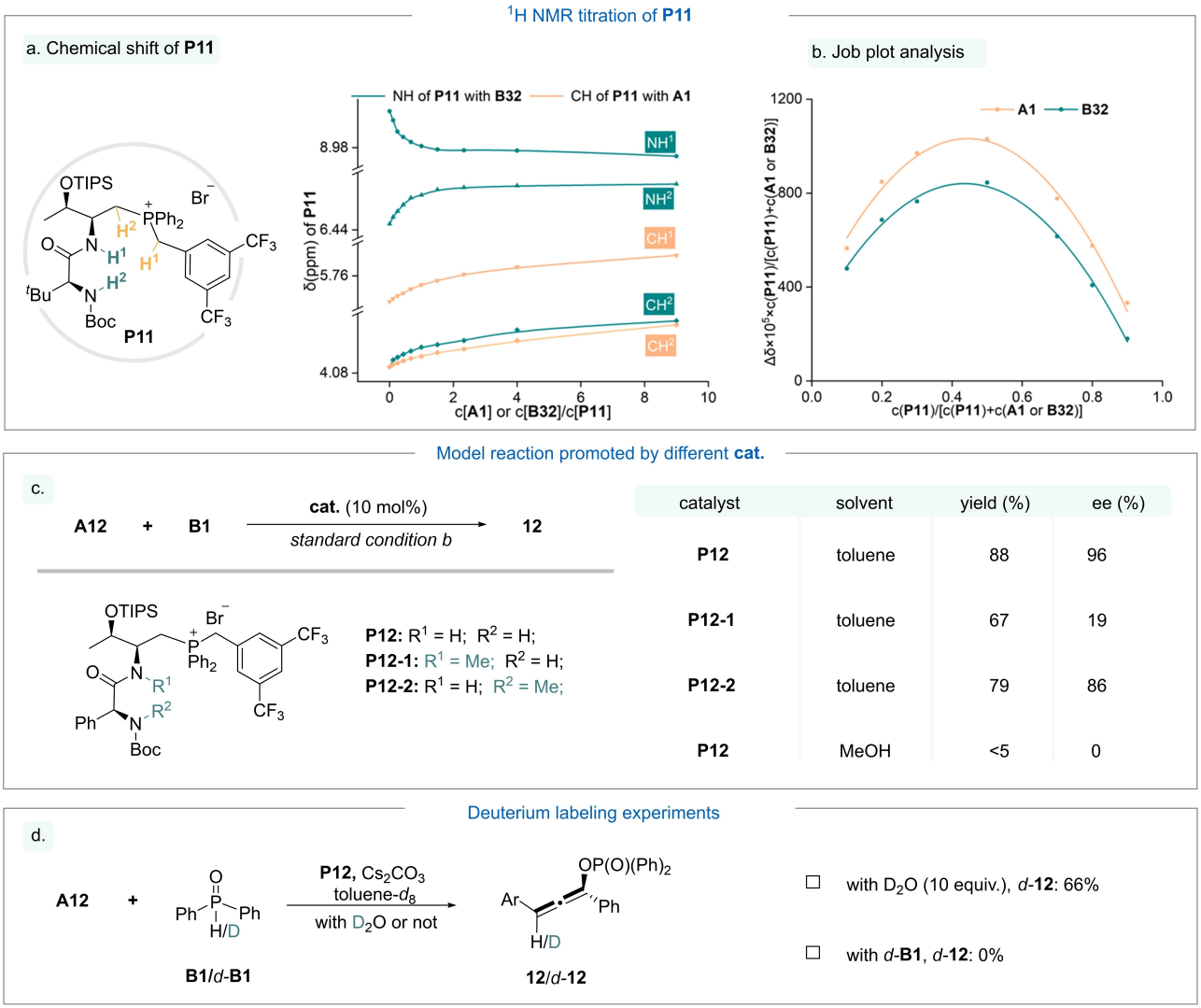


Figure 5. Mechanistic studies.

Conclusions

In summary, we have presented an unprecedented synergistic catalytic cascade protocol for the asymmetric synthesis of a range of chiral phosphorus allenes by peptide-mimic phosphonium salt (PPS) catalysis. This is achieved through a 1,2-addition reaction of ethynyl ketones with phosphine oxides, followed by catalyst-controlled stereospecific rearrangement and protonation, resulting in a wide variety of axially chiral allenyl phosphorus compounds in high yields with excellent stereoselectivities (up to 91 % yield and 96 % ee) under transition-metal-free conditions. Scaled-up preparation and synthetic transformation further demonstrate the broad applicability of this methodology. Furthermore, extensive mechanistic studies elaborate that the process involves an asymmetric 1,2-addition reaction as the enantio-determining step, followed by catalyst-facilitated stereospecific rearrangement and protonation, leading to a cascade process with central-to-axial chirality transfer. Under PPS catalysis, the 1,2-addition rather than the 1,4-addition occurs, confirming the strong catalytic capability of PPS catalysts. This opens up a more promising and viable strategy for effective transformation of ethynyl ketones, and further investigations into more challening reactions are currently underway in our laboratory.

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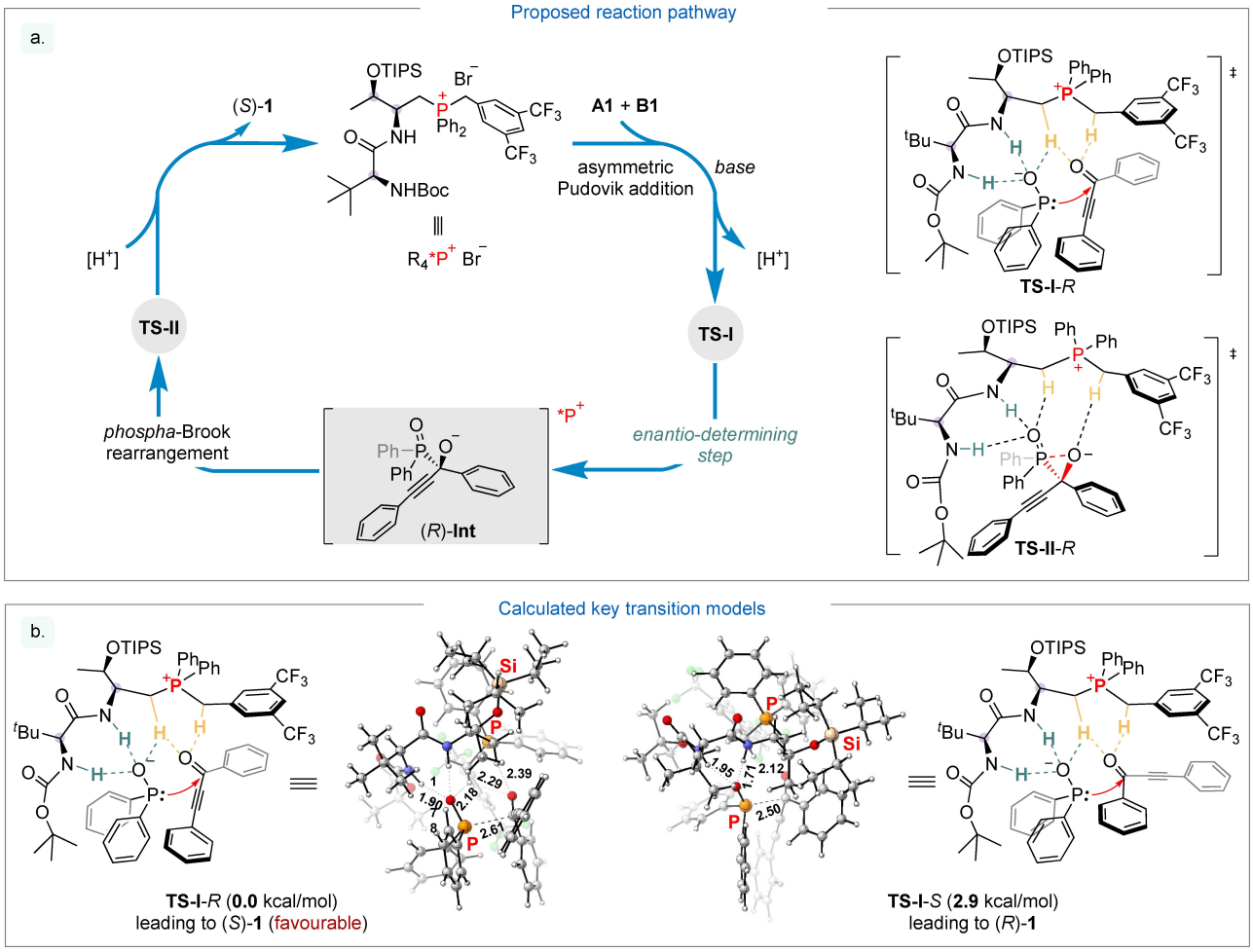


Figure 6. Proposed reaction pathway and calculated key transition models.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Peptide-mimic Catalysis · Axially Chiral Allenes · Cascade Reaction · Pudovik Addition · *Phospha*-Brook Rearrangement

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